

1 Firm's cost minimization with CES production function

2 Define:

X_i = the quantity of input i

W_i = the price of input i

Q = the quantity of output

δ_i = share parameter for input i

$-1 < \rho < \infty$ is parameter.

Firm's problem is to minimize cost of production subject to a given technology:

$$\min_{X_i} \sum_i W_i X_i \text{ s.t. } Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

$$\mathcal{L} = \sum_i W_i X_i + \lambda \left\{ Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} \right\}$$

First order conditions:

$$\frac{\partial \mathcal{L}}{\partial X_i} = 0 \text{ for all } i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

$$W_i - \lambda \left(-\frac{1}{\rho} \right) \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}-1} (-\rho \delta_i X_i^{-\rho-1}) = 0 \text{ for all } i = 1, \dots, n \quad (1)$$

$$Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} = 0 \quad (2)$$

Simplify 1:

$$W_i = \lambda \delta_i X_i^{-\rho-1} \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}-1} \text{ for all } i = 1, \dots, n$$

Ratio of two foc's (1):

$$\frac{W_i}{W_j} = \frac{\delta_i X_i^{-\rho-1}}{\delta_j X_j^{-\rho-1}} \text{ for all } i = 1, \dots, n \quad (3)$$

Solve 3 for X_i :

$$\begin{aligned} X_i^{-\rho-1} &= \frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \Leftrightarrow X_i = \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{-\frac{1}{\rho-1}} \\ X_i^{-\rho} &= \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{-\frac{\rho}{\rho-1}} \Leftrightarrow X_i^{-\rho} = \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{\rho}{1+\rho}} \end{aligned} \quad (4)$$

Substitute 4 into 2:

$$\begin{aligned} Q &= \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j X_j^{\rho+1}} \right)^{\frac{\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}} = \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} X_j^{\frac{-(\rho+1)\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}} \\ &= X_j \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{-\frac{1}{\rho}} \end{aligned}$$

Solve for X_j :

$$\begin{aligned} X_j &= Q \left\{ \sum_i \delta_i \left(\frac{\delta_j}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} \\ &= Q \left\{ \sum_i \delta_j^{\frac{\rho}{1+\rho}} \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \left\{ \delta_j^{\frac{\rho}{1+\rho}} \sum_i \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} \\ &= Q \delta_j^{\frac{1}{1+\rho}} \left\{ \sum_i \delta_i \left(\frac{1}{\delta_i} \frac{W_i}{W_j} \right)^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \delta_j^{\frac{1}{1+\rho}} \left\{ W_j^{-\frac{\rho}{1+\rho}} \sum_i \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} \\ &= Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \sum_i \delta_i \delta_i^{-\frac{\rho}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} = Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \sum_i \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right\}^{\frac{1}{\rho}} \\ &= Q \delta_j^{\frac{1}{1+\rho}} W_j^{-\frac{1}{1+\rho}} \left\{ \left(\sum_i \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \right\}^{\frac{1}{\rho} \frac{-\rho}{1+\rho}} = Q \delta_j^{\frac{1}{1+\rho}} \left\{ \frac{W_j}{\left(\sum_i \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}} \right\}^{-\frac{1}{1+\rho}} \\ &= Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}} \end{aligned} \quad (5)$$

where c :

$$c = \left(\sum_i \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} \quad (6)$$

It can be shown that c is the unit cost of production. Define C as the total cost:

$$\begin{aligned}
C &= \sum W_i X_i = \sum W_i Q \delta_i^{\frac{1}{1+\rho}} \left(\frac{W_i}{c} \right)^{-\frac{1}{1+\rho}} = Q c^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{1-\frac{1}{1+\rho}} \\
&= Q c^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{1+\rho-1}{1+\rho}} = Q c^{\frac{1}{1+\rho}} \sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \\
&= Q c^{\frac{1}{1+\rho}} \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho} \cdot \frac{\rho}{1+\rho}} = Q c^{\frac{1}{1+\rho}} c^{\frac{\rho}{1+\rho}} = Q c^{\frac{1+\rho}{1+\rho}} \\
C &= cQ
\end{aligned} \tag{7}$$

Define elasticity of substitution:

$$\begin{aligned}
\sigma &= \frac{1}{1+\rho}, \text{ or} \\
\sigma(1+\rho) &= 1 \Leftrightarrow \sigma + \sigma\rho = 1 \Leftrightarrow \sigma\rho = 1 - \sigma \\
\rho &= \frac{1 - \sigma}{\sigma}
\end{aligned} \tag{8}$$

Questions:

1. Interpret the elasticity of substitution!
2. What is its relationship with *factor intensity*?

Substitute: 8 into 5:

$$\begin{aligned}
X_j &= Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\rho}} = Q \delta_j^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} \left(\frac{W_j}{c} \right)^{-\frac{1}{1+\frac{1-\sigma}{\sigma}}} = Q \delta_j^{\frac{\sigma}{\sigma+1-\sigma}} \left(\frac{W_j}{c} \right)^{-\frac{1}{\sigma+1-\sigma}} \\
X_j &= Q \delta_j^{\sigma} \left(\frac{W_j}{c} \right)^{-\sigma}
\end{aligned} \tag{9}$$

Substitute: 8 into 6:

$$\begin{aligned}
c &= \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}} = \left(\sum \delta_i^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{1+\frac{1-\sigma}{\sigma}}} \right)^{\frac{1+\frac{1-\sigma}{\sigma}}{\frac{1-\sigma}{\sigma}}} \\
&= \left(\sum \delta_i^{\frac{\sigma}{\sigma+1-\sigma}} W_i^{\frac{\frac{1-\sigma}{\sigma}}{\frac{\sigma+1-\sigma}{\sigma}}} \right)^{\frac{\frac{\sigma+1-\sigma}{\sigma}}{\frac{1-\sigma}{\sigma}}} = \left(\sum \delta_i^{\frac{1}{\sigma}} W_i^{\frac{1-\sigma}{\frac{1}{\sigma}}} \right)^{\frac{\frac{1}{\sigma}}{\frac{1-\sigma}{\sigma}}} \\
c &= \left(\sum \delta_i^{\sigma} W_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}
\end{aligned} \tag{10}$$

Cost shares:

$$\begin{aligned}
S_j &= \frac{W_j X_j}{C} = \frac{W_j Q \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}}}{cQ} = \frac{W_j \delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}}}{c} \\
&= \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{1+\rho-1}{1+\rho}} c^{\frac{1}{1+\rho}}}{c} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{c c^{-\frac{1}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{c^{\frac{1+\rho-1}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{c^{\frac{\rho}{1+\rho}}} \\
&= \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{\left(\left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}}\right)^{\frac{\rho}{1+\rho}}} = \frac{\delta_j^{\frac{1}{1+\rho}} W_j^{\frac{\rho}{1+\rho}}}{\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}}
\end{aligned}$$

Cost share with σ :

$$\begin{aligned}
S_j &= \frac{\delta_j^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_j^{\frac{1-\sigma}{1+\frac{1-\sigma}{\sigma}}}}{\sum \delta_i^{\frac{1}{1+\frac{1-\sigma}{\sigma}}} W_i^{\frac{1-\sigma}{1+\frac{1-\sigma}{\sigma}}}} = \frac{\delta_j^{\frac{\frac{1-\sigma}{\sigma}+1-\sigma}{\sigma}} W_j^{\frac{\frac{1-\sigma}{\sigma}+1-\sigma}{\sigma}}}{\sum \delta_i^{\frac{\frac{1-\sigma}{\sigma}+1-\sigma}{\sigma}} W_i^{\frac{\frac{1-\sigma}{\sigma}+1-\sigma}{\sigma}}} = \frac{\delta_j^{\sigma} W_j^{1-\sigma}}{\sum \delta_i^{\sigma} W_i^{1-\sigma}} \\
S_j &= \frac{\delta_j^{\sigma} W_j^{1-\sigma}}{\sum \delta_i^{\sigma} W_i^{1-\sigma}}
\end{aligned}$$

Illustration with 2 factors production function:

$$Q = (\delta_K K^{-\rho} + \delta_L L^{-\rho})^{-\frac{1}{\rho}}$$

The unit cost:

$$c = (\delta_K^{\sigma} r^{1-\sigma} + \delta_L^{\sigma} w^{1-\sigma})^{\frac{1}{1-\sigma}}$$

The factor demands:

$$\begin{aligned}
K &= Q \delta_K^{\sigma} \left(\frac{r}{c}\right)^{-\sigma} \\
L &= Q \delta_L^{\sigma} \left(\frac{w}{c}\right)^{-\sigma}
\end{aligned}$$

Isoquant for illustration:

$$\begin{aligned}
Q &= (\delta_K K^{-\rho} + \delta_L L^{-\rho})^{-\frac{1}{\rho}} \Leftrightarrow Q^{-\rho} = \delta_K K^{-\rho} + \delta_L L^{-\rho} \\
\delta_K K^{-\rho} &= Q^{-\rho} - \delta_L L^{-\rho} \Leftrightarrow K^{-\rho} = \frac{Q^{-\rho} - \delta_L L^{-\rho}}{\delta_K} \\
K^Q(Q, L) &= \left(\frac{Q^{-\rho} - \delta_L L^{-\rho}}{\delta_K}\right)^{-\frac{1}{\rho}}, \text{ or} \\
K^Q(Q, L) &= \left(\frac{Q^{-\frac{1-\sigma}{\sigma}} - \delta_L L^{-\frac{1-\sigma}{\sigma}}}{\delta_K}\right)^{-\frac{1}{1-\sigma}} = \left(\frac{Q^{\frac{\sigma-1}{\sigma}} - \delta_L L^{\frac{\sigma-1}{\sigma}}}{\delta_K}\right)^{-\frac{\sigma}{1-\sigma}}
\end{aligned}$$

Isocost:

$$C = rK + wL \Leftrightarrow rK = C - wL \Leftrightarrow K^C(Q) = \frac{cQ - wL}{r}$$

Factor intensity:

$$f = \frac{Q\delta_K^\sigma \left(\frac{r}{c}\right)^{-\sigma}}{Q\delta_L^\sigma \left(\frac{w}{c}\right)^{-\sigma}} = \left(\frac{\delta_K}{\delta_L} \frac{w}{r}\right)^\sigma$$

Question:

1. What is *homotheticity*?
2. What is the connection between *homotheticity* and factor demand elasticity with respect to output?

3 Firm's cost minimization with CES production function

$$\min_{x_i} \sum_i w_i x_i \text{ s.t. } y = \prod_i x_i^{\alpha_i}$$

Define $k = \sum_i \alpha_i$ as returns to scale parameter.

$$k = \begin{cases} > 1 \Rightarrow IRS \\ = 1 \Rightarrow CRS \\ < 1 \Rightarrow DRS \end{cases}$$

Construct the Lagrange function:

$$\mathcal{L} = \sum_i w_i x_i + \lambda \left(y - \prod_i x_i^{\alpha_i} \right)$$

FOCs:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i} &= w_i - \lambda \alpha_i x_i^{\alpha_i-1} \prod_{j \neq i} x_j^{\alpha_j} = 0 \\ &= w_i - \lambda \alpha_i x_i^{-1} x_i^{\alpha_i} \prod_{j \neq i} x_j^{\alpha_j} = 0 \\ &= w_i - \lambda \frac{\alpha_i}{x_i} \prod_j x_j^{\alpha_j} = w_i + \lambda \frac{\alpha_i}{x_i} y = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_i} &= y - \prod_i x_i^{\alpha_i} = 0 \end{aligned}$$

$$\frac{w_i}{w_j} = \frac{\lambda \frac{\alpha_i}{x_i} y}{\lambda \frac{\alpha_j}{x_j} y} = \frac{\alpha_i}{x_i} \frac{x_j}{\alpha_j}$$

To find the solution:

$$\frac{w_i}{w_j} = \frac{\alpha_i}{x_i} \frac{x_j}{\alpha_j} \Rightarrow x_i = \frac{\alpha_i x_j w_j}{\alpha_j w_i}$$

Substitute into production function:

$$\begin{aligned}
y &= \prod_i x_i^{\alpha_i} = \prod_i \left(\frac{\alpha_i x_j w_j}{\alpha_j w_i} \right)^{\alpha_i} = \prod_i \left(\frac{x_j w_j}{\alpha_j} \right)^{\alpha_i} \prod_i \left(\frac{\alpha_i}{w_i} \right)^{\alpha_i} \\
y &= \left(\frac{x_j w_j}{\alpha_j} \right)^{\sum_i \alpha_i} \prod_i \left(\frac{\alpha_i}{w_i} \right)^{\alpha_i} = \left(\frac{x_j w_j}{\alpha_j} \right)^k \prod_i \left(\frac{\alpha_i}{w_i} \right)^{\alpha_i} \\
y^{\frac{1}{k}} &= \frac{x_j w_j}{\alpha_j} \left(\prod_i \left(\frac{\alpha_i}{w_i} \right)^{\alpha_i} \right)^{\frac{1}{k}} \\
x_j &= \frac{\alpha_j}{w_j} y^{\frac{1}{k}} \left(\prod_i \left(\frac{w_i}{\alpha_i} \right)^{\alpha_i} \right)^{\frac{1}{k}} \\
x_j^* &= \frac{\alpha_j c^{\frac{1}{k}} y^{\frac{1}{k}}}{w_j} \text{ where } c = \prod_i \left(\frac{w_i}{\alpha_i} \right)^{\alpha_i}
\end{aligned}$$

Cost function:

$$\begin{aligned}
C &= \sum_i w_i x_i^* = \sum_i w_i \frac{\alpha_i c^{\frac{1}{k}} y^{\frac{1}{k}}}{w_i} = \sum_i \alpha_i c^{\frac{1}{k}} y^{\frac{1}{k}} = c^{\frac{1}{k}} y^{\frac{1}{k}} \sum_i \alpha_i \\
C &= k c^{\frac{1}{k}} y^{\frac{1}{k}}
\end{aligned}$$

Marginal cost:

$$\frac{dC}{dy} = \frac{1}{k} k c^{\frac{1}{k}} y^{\frac{1}{k}-1} = c^{\frac{1}{k}} y^{\frac{1-k}{k}}$$

Average cost:

$$\frac{C}{y} = k c^{\frac{1}{k}} y^{\frac{1}{k}} y^{-1} = k c^{\frac{1}{k}} y^{\frac{1-k}{k}}$$